Algebra SEP Worksheet 10 8/18/22

Miscellaneous/Review

* **1.**

On this problem, no justification is required; only the answer will be graded.

(a) Give an example of a non-diagonalizable matrix in $\mathrm{GL}_3(\mathbb{F}_p)$ (for some p).

(b) Give an example of a nontrivial proper normal subgroup of $\mathrm{GL}_3(\mathbb{F}_p)$ (for some p).

(c) Give an example of a one-sided ideal (either left or right) in $M_2(\mathbb{C})$ which is not a two-sided ideal.

(d) Give an example of a $\mathbb{Z}$-module which is flat but not free.

(e) Give an example of an exact sequence of abelian groups which is not split.

Solution 1.

(a) Let $p = 2$ and consider $M = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

(b) Let $p = 3$. Then $\mathrm{SL}_3(\mathbb{F}_3)$ is a proper normal subgroup of $\mathrm{GL}_3(\mathbb{F}_3)$.

(c) Let I be the ideal of matrices of the form $\begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix}$. Then I is a left ideal but not a right ideal.

(d) $\mathbb{Q}$

(e) $0 \to \mathbb{Z} \to \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$

* **2.**

On this problem, no justification is required; only the answer will be graded.

(a) Give an example of a finite extension of $\mathbb{Q}$ which is not Galois.

(b) Give an example of a non-trivial, proper, two-sided ideal in $M_2(\mathbb{Z})$.

(c) Give an example of an injective map $M \to N$ of $k[x]$-modules (k a field) and another $k[x]$-module R such that $M \otimes R \to N \otimes R$ is not injective.

(d) Give an example of a commutative ring R and prime ideals p, q, q' in R such that the extension $\overline{q}$ of q to the localization R_p is prime, while the extension $\overline{q'}$ of q' to R_p is not prime.

Solution 2.

(a) $\mathbb{Q}(\sqrt[4]{2})$

(b) $M_2(2\mathbb{Z})$

(c) Consider $k[x] \xrightarrow{\cdot x} k[x]$ and $R = k[x]/(x)$. Then $k[x] \otimes_{k[x]} R \cong R \xrightarrow{\cdot x} k[x] \otimes_{k[x]} R \cong R$ is not injective since it is the zero map.

(d) Let $R = \mathbb{Z}$, $p = (2)$, $q = (0)$, and $q' = (3)$.

$*$ **3.**

On this problem, no justification is required; only the answer will be graded. Let S_3 be the symmetric group on 3 letters, and let R be the group ring $\mathbb{Z}[S_3]$.

(a) Write down a nonzero element of R which is a zero-divisor.

(b) Write down an element in the center of R which is not in $\mathbb{Z}$.

(c) Write down an R-module which is not a free module.

Solution 3.

(a) Let e be the identity in S_3 and let $\sigma = (12)$. Then $e + \sigma$ is a nonzero element of R that is a zero-divisor since $(e + \sigma)(e - \sigma) = 0$.

(b) Let $\tau = (123)$, and consider the element $x = e + \tau + \tau^2$. The element x clearly commutes with τ, so it is enough to check that x commutes with σ. We have

$$\begin{aligned}\sigma x &= \sigma + \sigma\tau + \sigma\tau^2 \\ &= \sigma + \tau^{-1}\sigma + \tau^{-2}\sigma \\ &= \sigma + \tau^2\sigma + \tau\sigma \\ &= x\sigma.\end{aligned}$$

(c) Consider $\mathbb{Z}^3$ as an R-module, where S_3 acts by permuting the entries. So for example, $(2\sigma + \tau)(a, b, c) = (2b + c, 3a, 2c + b)$. For any free R-module M of finite rank we have $\text{rank}_{\mathbb{Z}\text{-mod}}(M) = 6\ \text{rank}_{R\text{-mod}}(M)$, since R has rank 6 as a free $\mathbb{Z}$-module. However, $\mathbb{Z}^3$ has rank 3 as a $\mathbb{Z}$-module, which is not divisible by 6, so it cannot be a free R-module.

$*$ **4.**

Decide if the following statements are true. If so, provide a proof, if not, provide a counterexample. Let M, N be finitely generated modules over a commutative ring A. Let $f\colon M \to N$ be a homomorphism of A-modules, and let $g\colon B \to A$ be a homomorphism of commutative rings.

(a) "If f is injective, so is the induced morphism $M \otimes_A M \to N \otimes_A N$."

(b) "If f is surjective, so is the induced morphism $M \otimes_A M \to N \otimes_A N$."

(c) "If g is injective, so is the induced morphism $M \otimes_B M \to M \otimes_A M$."

Solution 4.

(a) We should suspect this is false because tensor products usually don't play well with injections. So we're going to need either M or N to be non-flat, but how do we figure out which one we need?

The morphism $M \otimes_A M \to N \otimes_A N$ factors as $M \otimes_A M \to M \otimes_A N \to N \otimes_A N$. If the composition were injective, then in particular the first map $\mathrm{id} \otimes f$ would also be injective.

However, we know $\mathrm{id} \otimes f$ need not be injective if M is not flat, so that gives us an avenue to look for a counterexample.

Here's one possible counterexample: let $A = \mathbb{Z}$, $M = \mathbb{Z}/2\mathbb{Z}$, $N = \mathbb{Z}/4\mathbb{Z}$, and $f\colon M \to N$ given by $f(1) = 2$, which is injective. Then we have that $M \otimes_A M \cong \mathbb{Z}/2\mathbb{Z}$ where the nonzero element is $1 \otimes 1$, $N \otimes_A N \cong \mathbb{Z}/4\mathbb{Z}$, but the induced map sends $1 \otimes 1$ to $2 \otimes 2 = 4 \otimes 1 = 0 \times 1$, so the induced map is the zero map.

(b) This one is true. Looking at the factorization from part (a), both $\mathrm{id} \otimes f$ and $f \otimes \mathrm{id}$ are surjective, since tensor products preserve surjections. But then the composition of two surjections is a surjection.

(c) This one is mega false. For example, let $A = \mathbb{C}$, $B = \mathbb{R}$, and $M = \mathbb{C}$. Then $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \to \mathbb{C} \otimes_{\mathbb{C}} \mathbb{C}$ is not injective, because the one on the left has real dimension 4 and the one on the right has real dimension 2. In fact, I think quite to the contrary that $M \otimes_B M \to M \otimes_A M$ is surjective when $B \to A$ is injective, and visa versa.

∗ **5.**

Let $K \subseteq F \subseteq E$ be fields with $E = F[\alpha]$ and with $\alpha^n \in F$ for some positive integer n. Suppose K contains a primitive nth root of unity, and let L be a field with $K \subseteq L \subseteq E$ and $L \cap F = K$.

(a) If L is Galois over K, show that $L = K[\beta]$ for some element β with $\beta^n \in K$.

(b) Show by example that L need not be Galois over K. (Hint. Take $n = 2$.)

Solution 5.

(a) As far as I can tell, this problem involves using a famous theorem by Kummer classifying certain cyclic extensions over fields containing an nth root of unity. If I find a better solution I'll update this worksheet.

(b) Let $K = \mathbb{Q}$, F and L be different fields containing cube roots of 2, say $F = \mathbb{Q}(\sqrt[3]{2})$ and $L = \mathbb{Q}(\zeta_3 \sqrt[3]{2})$, and $E = FL = \mathbb{Q}(\sqrt[3]{2}, \zeta_3)$. Then E is a quadratic extension of F, so it can be written as $F(\sqrt{\beta})$ (in fact $F(\sqrt{-3})$). However, L cannot be written as such because it is a degree 3 extension.

∗ **6.**

Consider the field with 11 elements $F = \mathbb{F}_{11}$. Let G denote the cyclic group of order 11 (written multiplicatively), with generator $r \in G$. Denote by FG the group algebra of G (also sometimes denoted $F[G]$). We consider r as an element of FG, and let $T\colon FG \to FG$ be the F-linear map given by $T(x) = rx$ for all $x \in FG$. Find the Jordan canonical form of T.

Solution 6.

We'll start by determining the minimal and characteristic polynomials of T. On the one hand, we know that $T^{11} = Id$ so the minimal polynomial $f(y)$ divides $y^{11} - 1$. On the other hand, if the degree of f is less than 11, then $f(T) : FG \to FG$ takes 1 to $f(r) \neq 0$ so $f(T) \neq 0$, so this tells us that $y^{11} - 1$ is the minimal polynomial and since it has degree 11 it is also the characteristic polynomial of T. In characteristic 11, $y^{11} - 1 = (y-1)^{11}$ so 1 is the only eigenvalue of T and the size of the largest Jordan block with eigenvalue 1 is 11, so the Jordan canonical form of T consists of a single 11×11 Jordan block with eigenvalue 1.

$*$ **7.**

Let F be a field of prime characteristic p. Suppose $E = F(\alpha)$ is a simple extension such that $\alpha \notin F$ but $\alpha^p - \alpha \in F$.

(a) Find $[E : F]$.

(b) Prove that E/F is a Galois extension.

(c) Find the Galois group $\mathrm{Gal}(E/F)$.

Hint: What is $(x+1)^p - (x+1)$ in characteristic p?

Solution 7.

(a) Let $\alpha^p - \alpha = b \in F$, so that α is a root of the polynomial $f(x) = x^p - x - b$. Thus, $[E : F] \leq p$. We will show that $[E : F] = p$, but we will have to come back to it.

(b) Notice that $(x+1)^p - (x+1) = x^p - x$, so $\alpha, \alpha+1, \ldots, \alpha+p-1$ are the p roots of f. Thus, f splits in E. Furthermore, f clearly has no repeated roots, so f is separable. Thus, E is the splitting field of a separable polynomial, and thus is Galois.

(c) The Galois group $\mathrm{Gal}(E/F)$ has size at most p, since we have already seen that $[E : F] \leq p$. But it is easy to check that $\alpha \mapsto \alpha + i$ gives p different automorphisms of $\mathrm{Gal}(E/F)$, so this must be all of them. Thus, $\mathrm{Gal}(E/F) \cong \mathbb{Z}/p\mathbb{Z}$, generated by $\alpha \mapsto \alpha + 1$.

(a) (again) Since E/F is a Galois extension and $|\mathrm{Gal}(E/F)| = p$, we must have $[E : F] = p$.

$*$ **8.**

Let k be a field, and let $R = M_2(k)$, the (noncommutative) ring of 2×2 matrices over k.

(a) Give an example of a simple left R-module M, and a simple right R-module N.

(b) Can you find another simple left R-module M' which is not isomorphic to M? (Either find one, or explain why there is none.)

(c) Compute $\dim_k N \otimes_R M$.

Solution 8.

(a) Let M be the vector space of 2×1 column vectors, and let N be the vector space of 1×2 row vectors.

(b) Suppose we have a simple left R-module M'. We will show that M' is isomorphic to M. For any nonzero $v \in M'$, we have that Rv' is a nontrivial submodule of M', hence must be all of M'. Fix a nonzero $v \in M'$, and define a map $M' \to M$ by $Av \mapsto A\begin{pmatrix}1 & 0\end{pmatrix}$, for $A \in R$. (Remember, this is well defined since $Rv = M'$.) This map is clearly onto. Notice now that its kernel is a submodule of M', hence is either 0 or M'. Since this map is not the zero map, the kernel must be 0, and hence this is an isomorphism between M' and M. (There's actually some cool afterlife to this fact: look up "Morita equivalence".)

(c) As noted above, any vector in M can be written as $A\begin{pmatrix}1 & 0\end{pmatrix}$, thus we can rewrite any simple tensor $u \otimes v$ as $uA \otimes \begin{pmatrix}1 & 0\end{pmatrix}$. Thus, every tensor in $N \otimes_R M$ can be written as $u \otimes \begin{pmatrix}1 & 0\end{pmatrix}$ for some $u \in N$. It is not hard to check that this gives an isomorphism of vector spaces $N \otimes_R M \cong N$, and thus $\dim_k N \otimes_R M = 2$.

$*$ **9.**

Let V be a non-zero finite dimensional vector space over $\mathbb{C}$. Given a linear transformation $A\colon V \to V$, show that the following are equivalent.

(i) There exists a linear transformation $P\colon V \to V$ such that $P^2 = I$ and $AP = -PA$.

(ii) There exists an invertible linear transformation $P\colon V \to V$ such that $AP = -PA$.

(iii) There exists a direct sum decomposition $V = V_1 \oplus V_2$ such that $AV_1 \subset V_2$ and $AV_2 \subset V_1$.

(Hint: Thinking about (iii) $\Rightarrow$ (i) helped me to think about (ii) $\Rightarrow$ (iii).)

Solution 9.

The implication (i) $\Rightarrow$ (ii) is immediate.

For (ii) $\Rightarrow$ (iii), we first begin by showing that for every eigenvalue λ of AP, $-\lambda$ is also an eigenvalue. Suppose $\lambda \neq 0$ is an eigenvalue of AP, and v is an eigenvector for λ. Then notice that $APAv = -A(AP)v = -\lambda Av$. Furthermore, we have that $Av = -P^{-1}APv = -\lambda P^{-1}v \neq 0$, so Av is an eigenvector for AP of eigenvalue $-\lambda$.

Now, divide the eigenvalues of AP into two sets, S_1 and S_2, where if a nonzero $\lambda \in S_1$, then $-\lambda \in S_2$, and visa versa. Let V_1 be the union of the λ-generalized eigenspaces for all $\lambda \in S_1$, and V_2 that for those $\lambda \in S_2$. Then our calculation in the previous paragraph shows that $AV_1 \subset V_2$ and $AV_2 \subset V_1$, as desired.

Finally, for (iii) $\Rightarrow$ (i), let P be given by the block diagonal matrix $\begin{pmatrix}-I & 0 \\ 0 & I\end{pmatrix}$. It is easy to check this matrix satisfies the properties we desire.